Enable the Network Encryption

VPN connection or Oracle Net encryption is also required for encryption-in-flight for any other database payload (e.g. data file or redo headers) that are not encrypted by TDE. In this lab, you use public internet to connect between on-premise and the cloud, so you need to enable the network encryption.

1. Work as oracle user, connect to the database as sysdba.

connect / **as** sysdba

2. Check the network service banner before encryption.

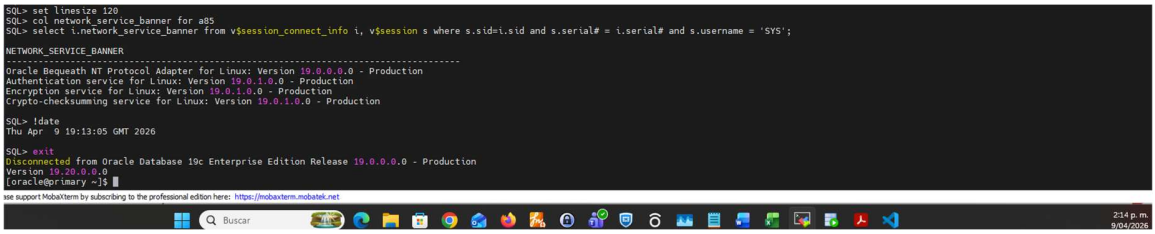
set linesize 120
col network_service_banner **for** a85
select i.network_service_banner **from** v$session_connect_info i, v$session s **where** s.sid=i.sid **and** s.serial# = i.serial# **and** s.username = 'SYS';

NETWORK_SERVICE_BANNER

Oracle Bequeath NT Protocol Adapter **for** Linux: Version 19.0.0.0.0 - Production
Authentication service **for** Linux: Version 19.0.0.0.0 - Production
Encryption service **for** Linux: Version 19.0.0.0.0 - Production
Crypto-checksumming service **for** Linux: Version 19.0.0.0.0 - Production

SQL> exit
Disconnected **from** Oracle Database 19c Enterprise Edition **Release** 19.0.0.0.0 - Production
Version 19.10.0.0.0
[oracle@primary ~]$

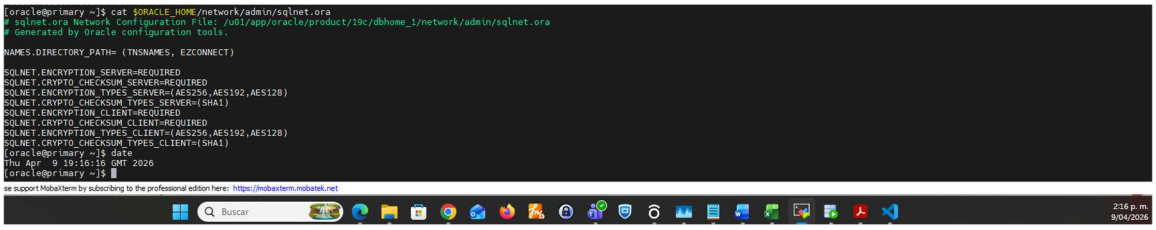
From the primary:



3. Edit the sqlnet.ora file.

vi $ORACLE_HOME/network/admin/sqlnet.ora
Add following lines in the sqlnet.ora, save and exit.
SQLNET.ENCRYPTION_SERVER=REQUIRED
SQLNET.CRYPTO_CHECKSUM_SERVER=REQUIRED
SQLNET.ENCRYPTION_TYPES_SERVER=(AES256,AES192,AES128)
SQLNET.CRYPTO_CHECKSUM_TYPES_SERVER=(SHA1)
SQLNET.ENCRYPTION_CLIENT=REQUIRED
SQLNET.CRYPTO_CHECKSUM_CLIENT=REQUIRED
SQLNET.ENCRYPTION_TYPES_CLIENT=(AES256,AES192,AES128)
SQLNET.CRYPTO_CHECKSUM_TYPES_CLIENT=(SHA1)

From the primary:



4. Check the network service banner again, the network encryption is enable now.

[oracle@workshop ~]$ sqlplus / **as** sysdba

SQL*Plus: **Release** 19.0.0.0.0 - Production **on** Fri Jan 31 03:51:46 2020
Version 19.10.0.0.0

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Connected **to**:
Oracle Database 19c Enterprise Edition **Release** 19.0.0.0.0 - Production
Version 19.10.0.0.0

set linesize 120
col network_service_banner **for** a85
select i.network_service_banner **from** v$session_connect_info i, v$session s **where** s.sid=i.sid **and** s.serial# = i.serial# **and** s.username = 'SYS';

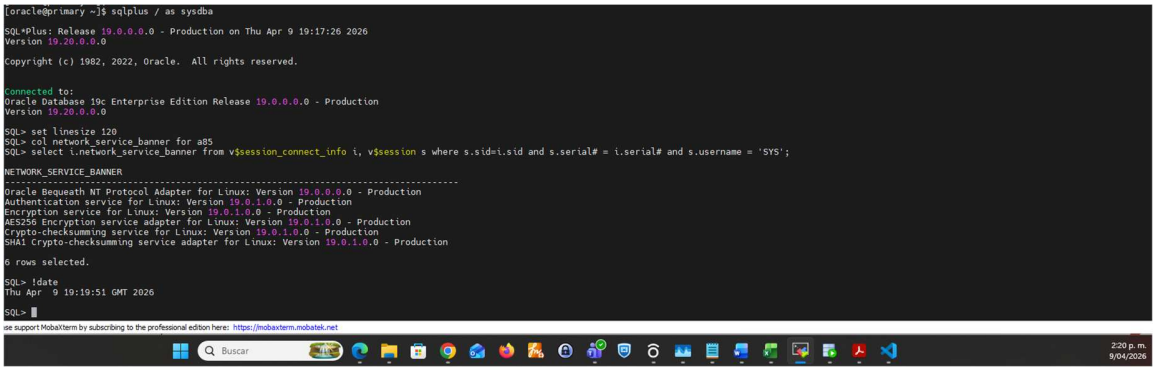
NETWORK_SERVICE_BANNER

Oracle Bequeath NT Protocol Adapter **for** Linux: Version 19.0.0.0.0 - Production
Authentication service **for** Linux: Version 19.0.0.0.0 - Production
Encryption service **for** Linux: Version 19.0.0.0.0 - Production
AES256 Encryption service adapter **for** Linux: Version 19.0.0.0.0 - Production
Crypto-checksumming service **for** Linux: Version 19.0.0.0.0 - Production
SHA1 Crypto-checksumming service adapter **for** Linux: Version 19.0.0.0.0 - Production

6 **rows** selected.

SQL>

From the primary:



```
[oracle@primary ~]$ sqlplus / as sysdba

SQL*Plus: Release 19.0.0.0.0 - Production on Thu Apr 9 19:17:26 2026
Version 19.20.0.0.0

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Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.20.0.0.0

SQL> set linesize 120
SQL> col network_service_banner for a85
SQL> select i.network_service_banner from v$session_connect_info i, v$session s where s.sid=i.sid and s.serial# = i.serial# and s.username = 'SYS';

NETWORK_SERVICE_BANNER
-------------------------------------------------------------------------------
Oracle Bequeath NT Protocol Adapter for Linux: Version 19.0.0.0.0 - Production
Authentication service for Linux: Version 19.0.1.0.0 - Production
Encryption service for Linux: Version 19.0.1.0.0 - Production
AES256 Encryption service adapter for Linux: Version 19.0.1.0.0 - Production
Crypto-checksumming service for Linux: Version 19.0.1.0.0 - Production
SHA1 Crypto-checksumming service adapter for Linux: Version 19.0.1.0.0 - Production

6 rows selected.

SQL> !date
Thu Apr  9 19:19:51 GMT 2026

SQL>
```